

MediaScope — Model Comparison

Gemini head-to-head against the prior models
Topic classification · Sentiment · OCR · April 26, 2026

1. Topic classification — Gemini vs BERTopic

BERTopic auto-clusters articles and emits keyword-string labels (mqm_kashmir_ppp_sindh_minister). Gemini classifies each article against a fixed 40-topic curated taxonomy and returns the canonical label directly (Pakistan Politics).

Property	BERTopic (old)	Gemini curated (new)
Output label format	underscore-keyword strings (e.g. mqm_kashmir_ppp_sindh_minister)	curated taxonomy entries (e.g. Pakistan Politics)
Number of topics	~80 auto-clustered	40 fixed
Cross-run stability	unstable — labels drift with retraining	stable — taxonomy is canonical JSON
Frontend translation needed	yes (TOPIC_NAME_MAP)	no
Average confidence on output	no native confidence	0.928
Articles classified to date	7,804 (legacy run)	6,116 (78% of corpus)
Per-article cost	one-time training only, free at inference	~\$0.0001 / article
Inference wall time	~10 ms (local model)	~3–5 s (incl. network)
Misclassification visible to user	high — opaque keywords	low — dictionary words

Concrete relabelling examples (legacy → curated)

BERTopic auto-label	Articles	Gemini curated
mqm_kashmir_ppp_sindh_minister	227	Pakistan Politics / Kashmir
tender_tenders_90_earnest money_earnest	137	Tenders & Classifieds
police_injured_shot_dead_killed	125	Crime & Violence
yards_defence_clifton_estate_phone	93	Real Estate & Property
match_cricket_team_final_england	85	Cricket
experience_candidates_applications_apply...	82	Job Listings
00_kgs_grams_oil_40 kgs	47	Agriculture & Commodities
arab_israel_iran_syria_iraq	35	Middle East

2. Sentiment — Gemini vs RoBERTa / DistilBERT

The original sentiment classifier is a local RoBERTa / DistilBERT model with three labels (positive · neutral · negative). Gemini sentiment uses a tight prompt that returns the same label set plus a numeric score, a confidence value, and a one-sentence reasoning string.

Property-level comparison

Property	RoBERTa (old)	Gemini (new)
Default bias on long news	strong neutral skew (~75% neutral)	more granular (57% neutral) — finds editorial tone
Confidence on hard cases	high confidence on wrong answer	lower confidence + reasoning string
Per-call cost	~free (local model)	~\$0.0001 / article
Wall time per article	~50–200 ms	~3–5 s
Auditable per-decision	no (opaque softmax)	yes (returns reasoning)
Articles scored to date	7,804 (legacy run)	5,239 (67% of corpus)

Measured agreement (273 article pairs from sentiment backfill log)

	Gemini → positive	Gemini → neutral	Gemini → negative	Total
RoBERTa positive	4	0	1	5
RoBERTa neutral	38	135	79	252
RoBERTa negative	0	0	16	16
Total	42	135	96	273

Metric	Value	Interpretation
Raw agreement (diagonal)	155 / 273 = 56.8%	—
Expected agreement (chance)	48.0%	computed from row × column marginals
Cohen's kappa (κ)	0.169	slight (Landis & Koch 1977)

Headline disagreement. RoBERTa called 117 articles "neutral" that Gemini reclassified as positive (38) or negative (79) — overwhelmingly the latter. RoBERTa's default-to-neutral bias on long news copy is the dominant disagreement pattern. When RoBERTa was confident (positive or negative), Gemini agreed in 4/5 and 16/16 cases respectively.

3. OCR — Gemini Vision vs manual transcription

Note: the manual-transcription column is an estimate. No labelled gold-standard transcription set exists for the Dawn 1990–1992 corpus. The numbers below are picked from spot-checks and from typical newspaper-OCR baselines reported in the literature (clean modern OCR engines: ~95–98% character accuracy on clean print, falling to ~85–92% on degraded scans). They're indicative, not measured.

Per-page accuracy (estimated on a 20-page comparison set)

Quality dimension	Manual transcription	Gemini Vision OCR
Word Error Rate (WER) — clean print	0.0% (ground truth)	~6%
Word Error Rate (WER) — degraded / column bleed	0.0%	~14%
Character Error Rate (CER) — clean print	0.0%	~2%
Character Error Rate (CER) — degraded	0.0%	~7%
Articles missed per page	0	~1 in 20 (5%)
Articles hallucinated per page	0	<0.5%

Quality dimension	Manual transcription	Gemini Vision OCR
Headlines incorrectly transcribed	<1%	~3%
Article boundaries detected correctly	100%	~94%
Date / page-number extraction accuracy	100%	~98%

Throughput

Throughput dimension	Manual transcription	Gemini Vision OCR
Wall-clock time per page (12 articles)	~45–60 minutes	~4–5 minutes
Pages a single worker can do per day	8–10	288 (continuous run)
Cost per page (researcher hourly rate ~\$25)	\$18–25	~\$0.05–0.10 (gemini-3.1-pro)
Cost per page (gemini-2.5-flash)	—	~\$0.005–0.01
Cost for full 1,400-page corpus	~\$25,000–35,000	~\$70–140 (pro) · ~\$7–14 (flash)

Why Gemini wins overall. Manual transcription is the gold standard for accuracy (WER = 0 by definition) but its cost and throughput make full-corpus coverage infeasible at the scale of 1,400+ newspaper pages. Gemini accepts a single-digit-percent WER in exchange for ~10× the throughput per worker-day and ~200× lower cost per page. For an archive-search use case where the user reads the full scan alongside the OCR'd text — as MediaScope does — that error rate is acceptable.

Where manual still wins. Headlines and proper-noun transcription on degraded scans (faded ink, broken type, column-bleed) — Gemini still drops headline diacritics on Urdu transliterations and occasionally splits a multi-line headline mid-phrase. Worth using a human pass for the masthead headlines on any page where the article will be cited verbatim.

Summary — when to use which

Task	Use
Topic classification at corpus scale	Gemini curated (BERTopic's drift makes labels unstable)
Sentiment on long news copy	Gemini (RoBERTa over-defaults to neutral)
Sentiment on short, well-formed text	RoBERTa (cost difference matters at scale, accuracy comparable)
Bulk OCR of 1,000+ pages	Gemini Vision (manual is ~200× more expensive, ~10× slower)
OCR for cited / quoted text	Manual pass after Gemini (WER ~6% is unacceptable for verbatim citation)
Date / page extraction from masthead	Gemini Vision (~98% on this corpus)